

NBS Foundations

CUNY EES 79903

Seminar on Nature-Based Solutions

Brett Branco and Kyle McKay

September 15, 2026

Lecture Overview:

- 415-430 – Housekeeping: field trip planning, projects
- 430-445 – Lecture: hybrid infrastructure
- 445-500 – Example: living breakwaters of Staten Island
- 500-615 – Discussion: Cohen-Shacham et al. (2019), Singhvi et al. (2022), Silliman et al. (2026)

	Leader #1	Leader #2
Sep 8: NBS fundamentals	Grace	Jesse
Sep 15: NBS fundamentals	Michelle	Shahela
Sep 22: Structured decision making and community engagement	Jesse	Aurelia
Sep 29: Analysis – ecological	Carol	Julia
Oct 6: Analysis – co-benefits	Shahela	Aurelia
Oct 20: Analysis – engr performance	Grace	Michelle
Oct 27: Analysis – engr performance	Abdellah	Julia
Nov 3: Alternatives analysis	Aurelia	Abdellah
Nov 10: Performance-based design	Aaryan	Michelle
Nov 17: Constructability	Grace	Aaryan
Nov 24: Performance monitoring	Shahela	Jesse
Dec 1: Performance monitoring	Julia	Carol
Dec 8: Life cycle analysis	Aaryan	Abdellah
Dec 15: Course synthesis	Kyle	Brett

Field trip planning

- October 11th – timing?
- Potential locations:
 - Prospect Park: stormwater walk
 - Manhattan: East Side Coastal Resiliency, Battery Park, Gansevoort
 - *Jamaica Bay: Shirley Chis olm, Rockaways
- Rough Agenda?

Preliminary Project Sign-ups (**submit your top 2**)

Topic	First option	Second option
Qualitative foundations of project planning	Jesse	Grace
Facilitation, engagement, and teaming		Jesse
Ecological analysis of NBS	Julia	
Economic analysis of NBS		
Co-benefits analysis of NBS		
Engineering analysis of NBS		Michelle
Alternatives analysis and structured decision making		
Performance-based design	Aaryan	
Constructability	Michelle	Aaryan
Performance monitoring		Julia
Project establishment		
Asset management	Grace	

What is hybrid infrastructure?

Taken liberally from:

McKay, Bledsoe, Aliu, and Webb. 2023. Accelerating Use of Hybrid Infrastructure Systems in Water Resources. ASCE Inspire 2023, Alexandria, Virginia.

A huge spectrum of natural to conventional infrastructure

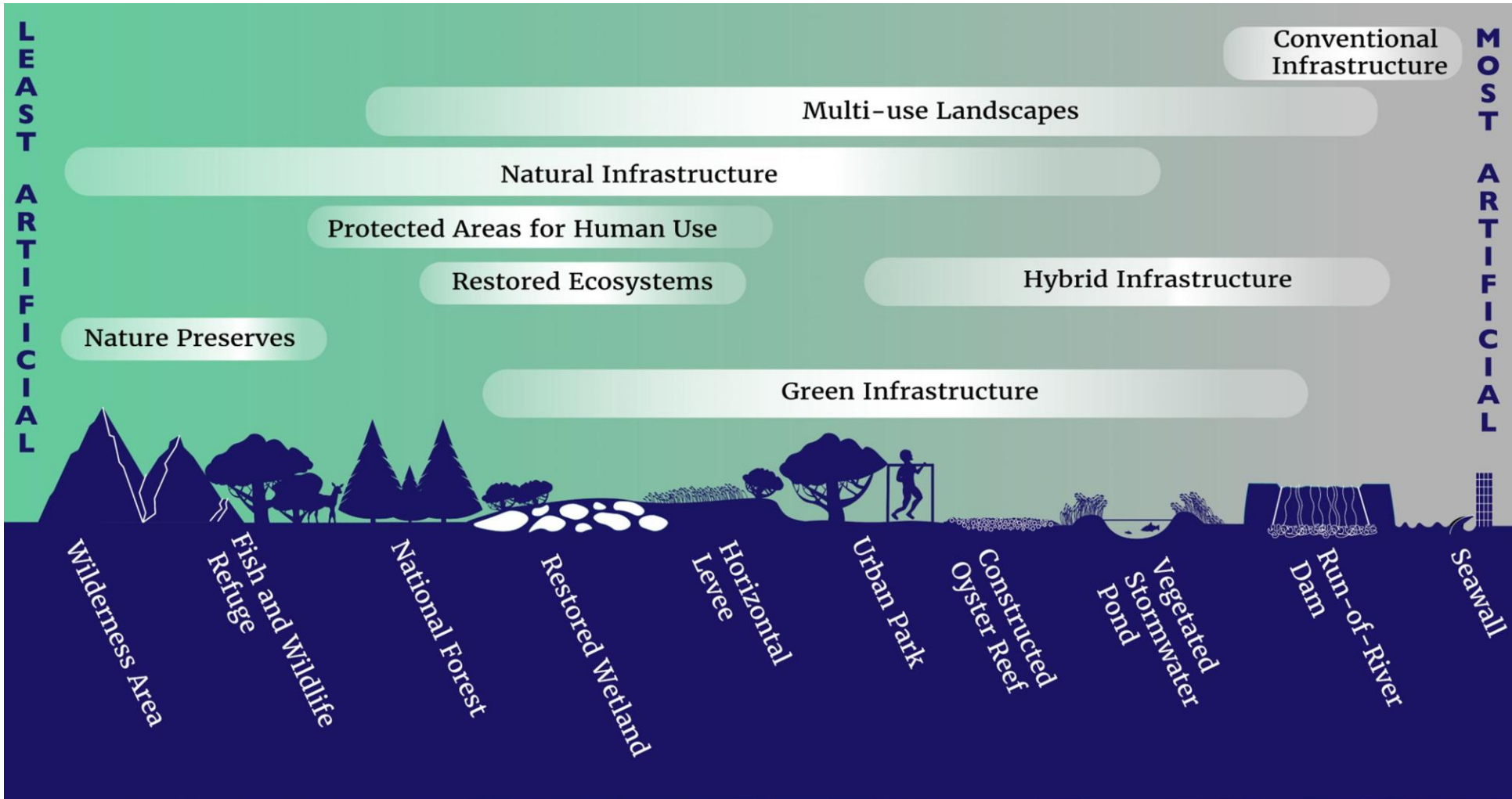


Figure: van Rees et al. (2023, PNAS)

The dialog often focuses on green or grey, but hybrid systems are more likely

How do we offer practical advice *for engineers and designers* that:

- Merges the best parts of green and grey?
- Accelerates the use of hybrid approaches?

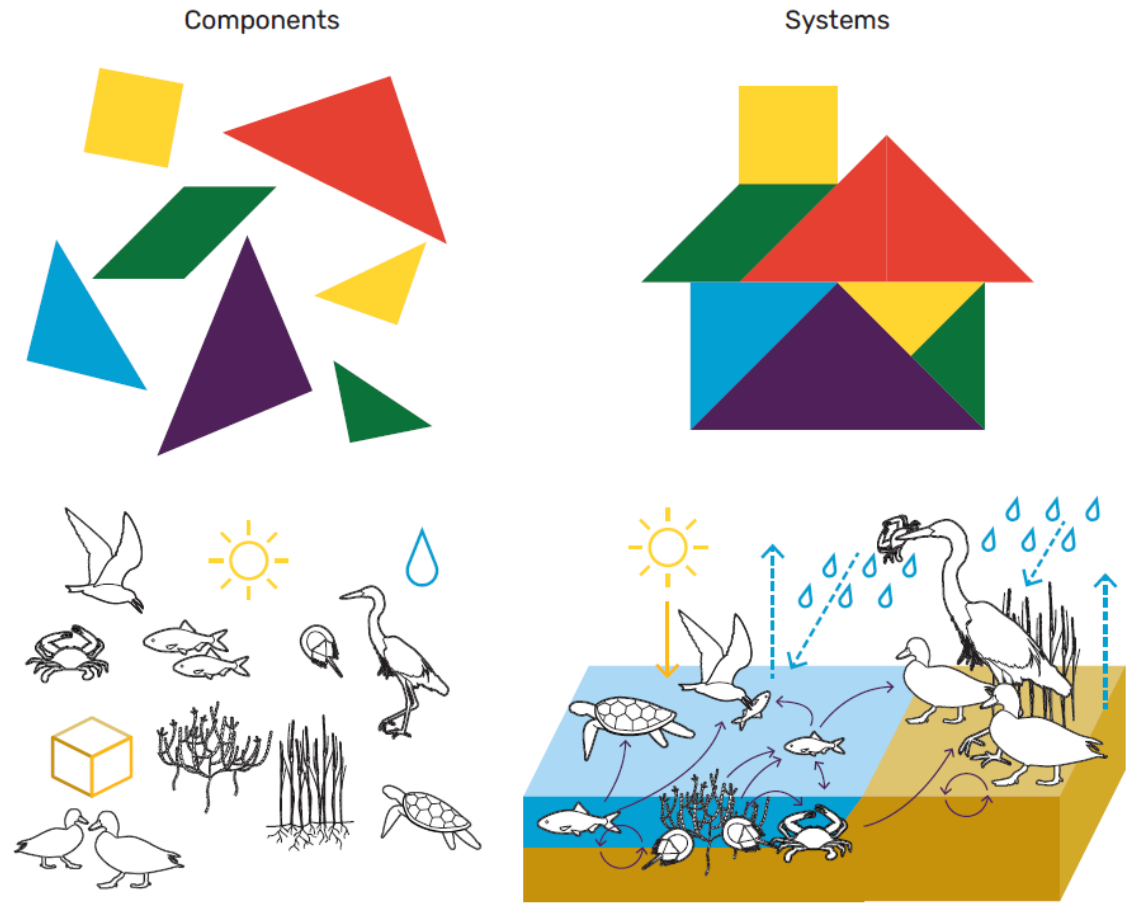
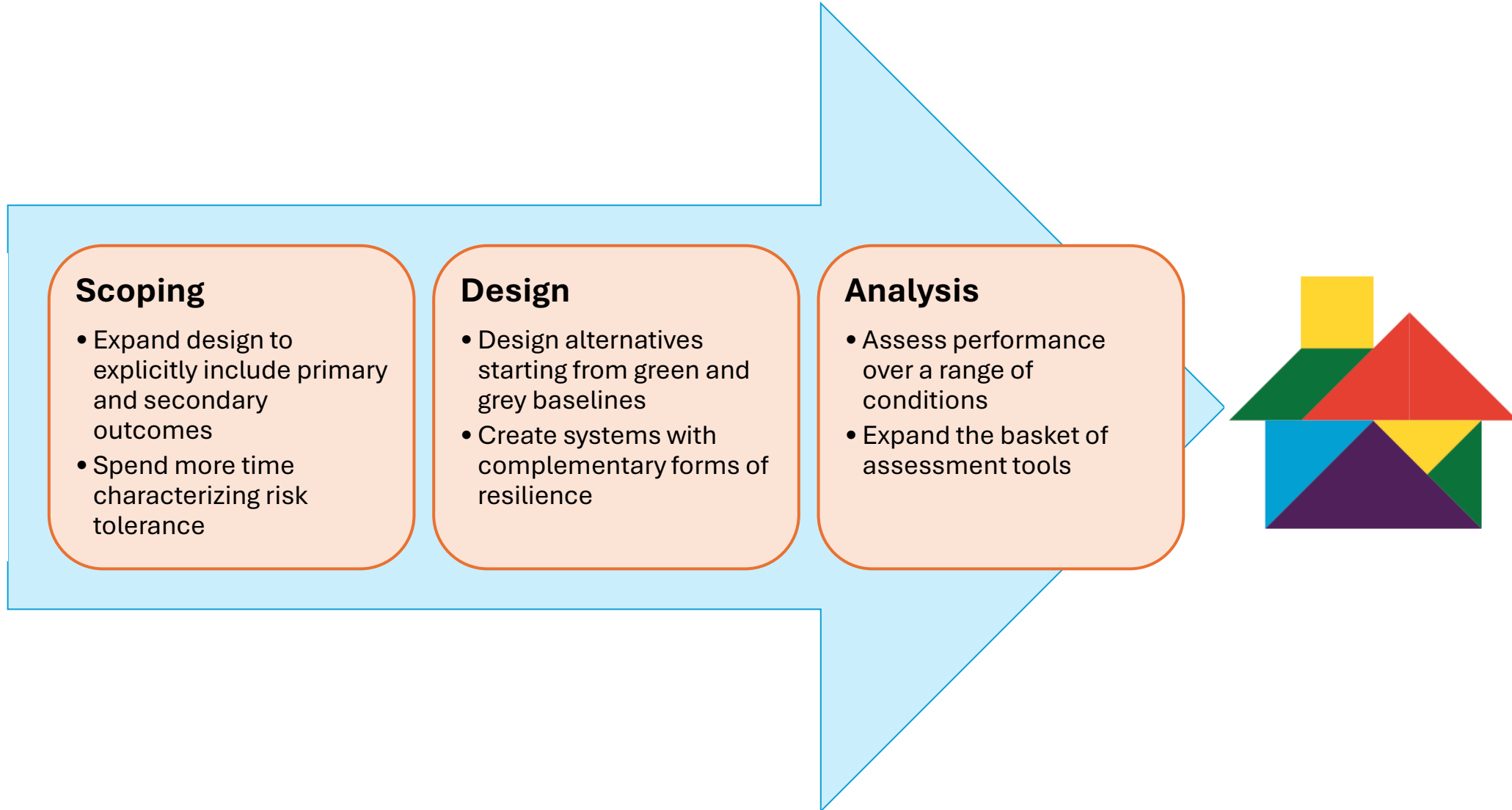


Figure: de Vries et al. (2021, Chapter 4 of the International Guidelines for NNBF)

Pragmatic actions to accelerate the use of hybrid green-grey systems



Expand design to explicitly include primary and secondary outcomes

Designing for a primary purpose does not preclude consideration of other outcomes.

- List primary and secondary outcomes
- Qualitatively rank the relative importance of outcomes
- Identify metrics for select outcomes.
- Consider how small-scale features could be added
- Explore unintended consequences

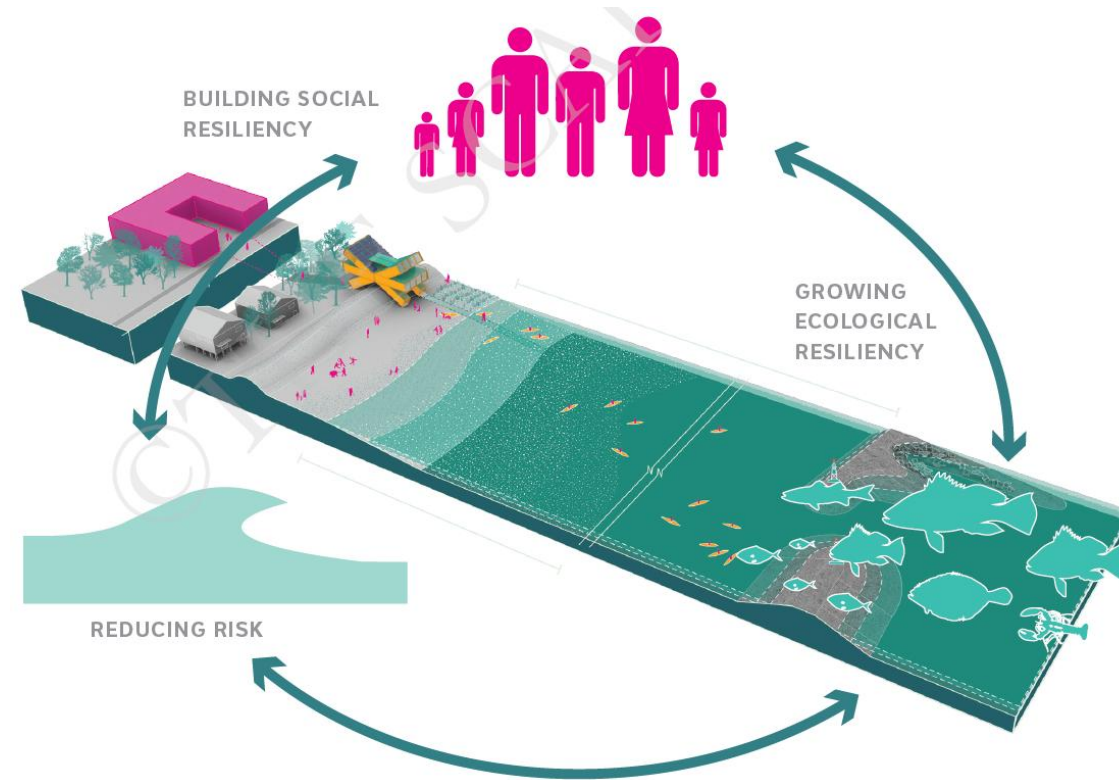


Figure: Conceptual model from Staten Island's "Living Breakwaters" (SCAPE 2014)

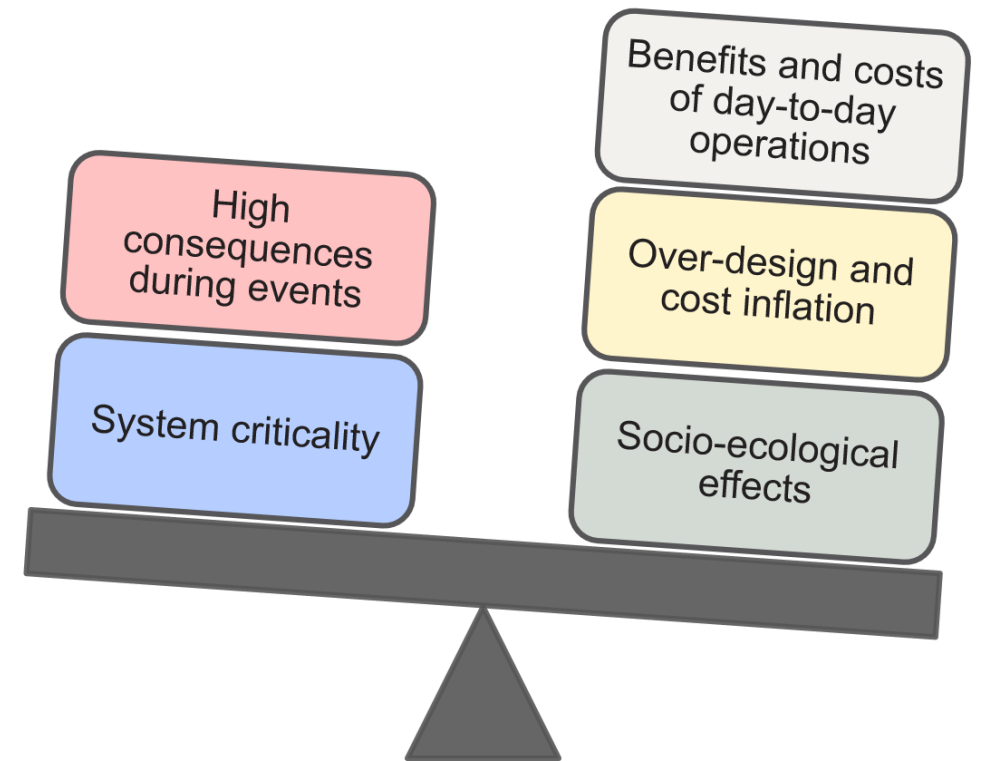
Spend more time characterizing risk tolerance

Not every system should be designed for same level of performance

- The relative risks of failure are crucial to choosing the right combination of methods
- Acknowledging uncertainties is not a weakness, but a practical reality
- Risk tolerances vary at scale of individuals, organizations, and contexts

Build it stout!

Safe-to-fail



Design alternatives starting from green and grey baselines

Don't assume any approach is preferred. Neither is a “condiment” for the other. Design in both directions.

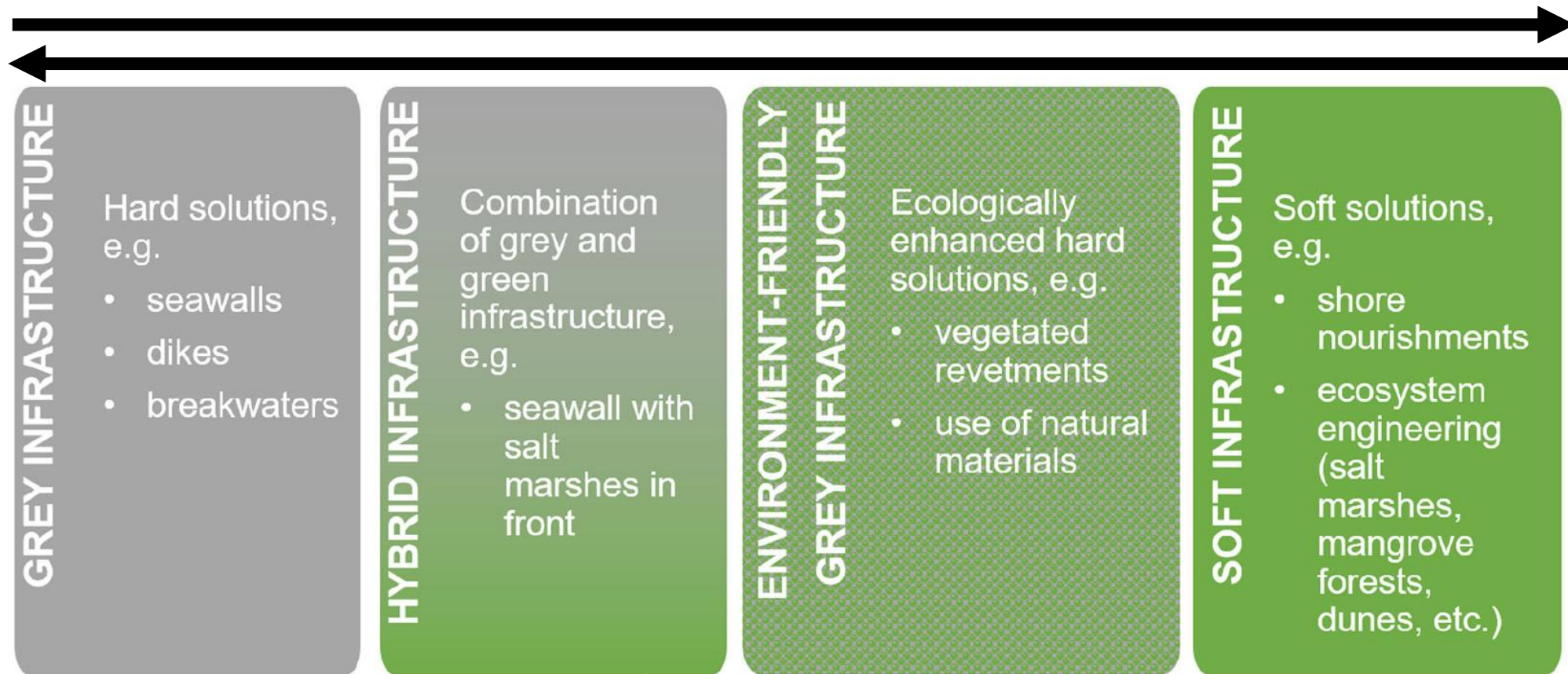


Figure: Schoonees et al. (2019, *Estuaries and Coasts*)

Create systems with complementary forms of resilience

Different forms of resilience can be an asset in design.

- Coupling grey and green features with different forms of resilience (prepare, absorb, recovery, adapt)
- Emphasizing the roles of adaptability and reversibility (rather than just recovery)
- Designing with multiple hazards in mind (e.g., flood vs. fire)
- Designing around sequences of events (e.g., Katrina-Rita, fire-then-flood,...)

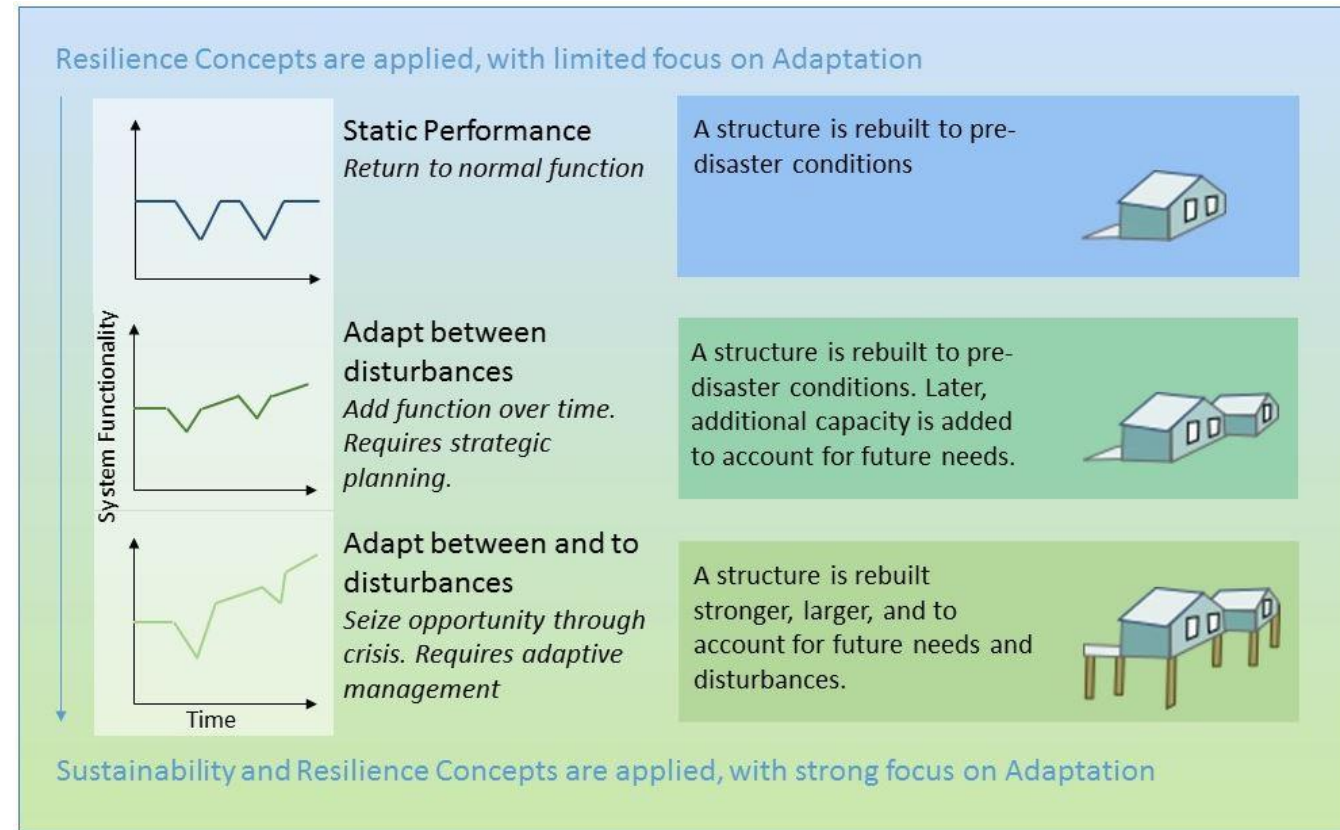


Figure: Chambers et al. (2019, ERDC/TN EMRRP-SR-86)

Assess performance over a range of conditions

Performance involves primary and secondary outcomes and assessment can begin with qualitative understanding.

- Are there gaps in performance for intermediate disturbances?
- What are the limits of both systems beyond the design thresholds?
- How do event magnitudes intersection with frequency through time?

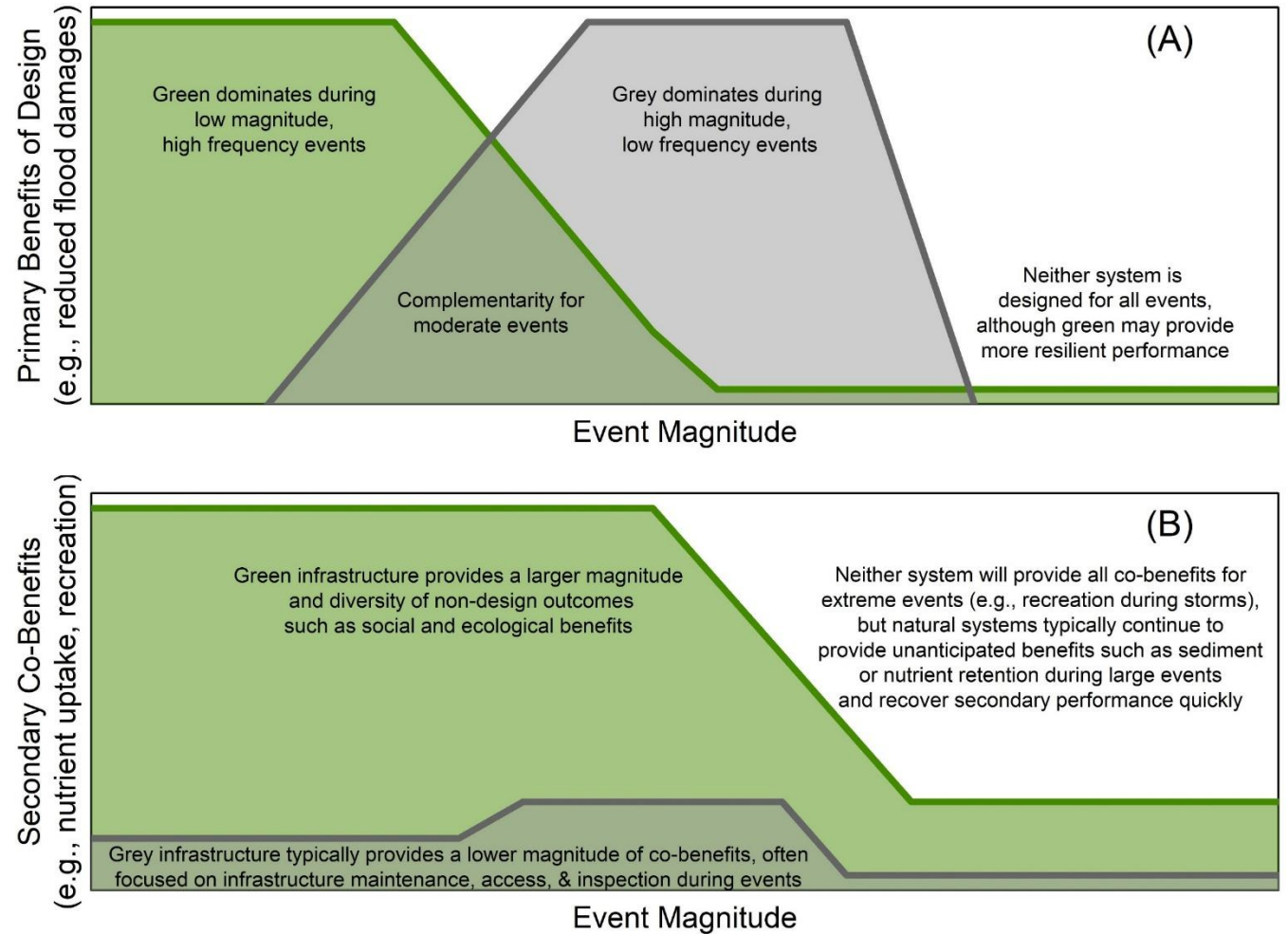
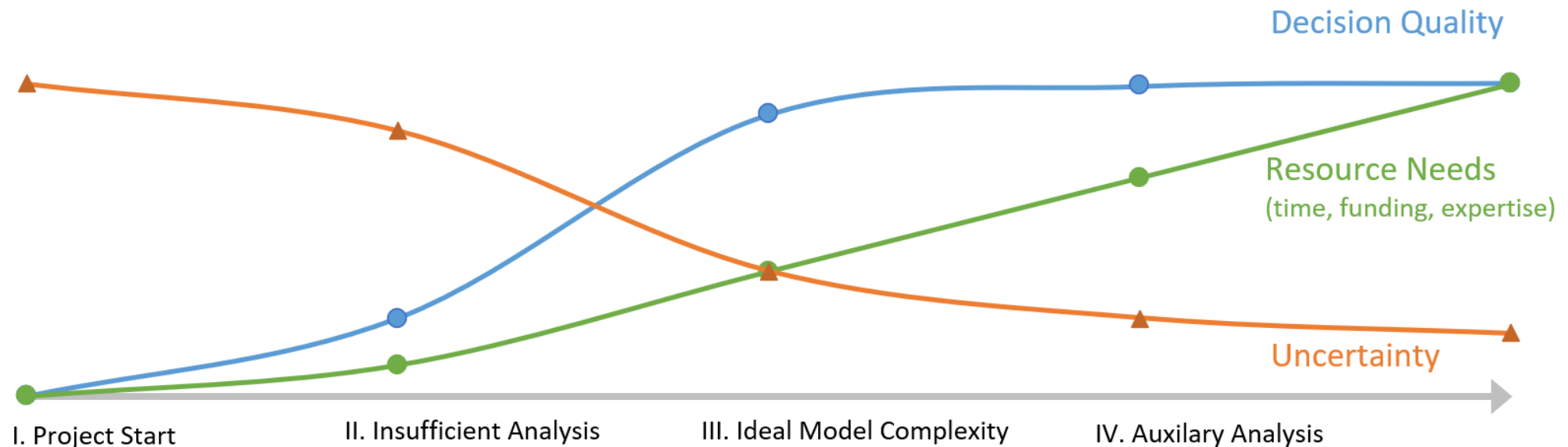


Figure: McKay et al. (*forthcoming*)

Expand the basket of assessment tools

“Telling the story” of hybrid systems may require more or different analyses:

- Broadening the use of standard tools
- Multi-disciplinary analysis of hybrid systems is hard, but worth it
- Additional communication may be required with decision-makers



Enabling conditions for executing hybrid infrastructure systems

- Teaming effectively to create truly interdisciplinary designs
- Creating shared visions across organizational missions
- Fostering collaborative relationships
- Leveraging funding for collective action
- Adequately funding and allocating time for process
- ...



Figure: Jamaica Bay, New York City (Baron and Wepler, USACE New York District)

NBS Example: Living Breakwaters

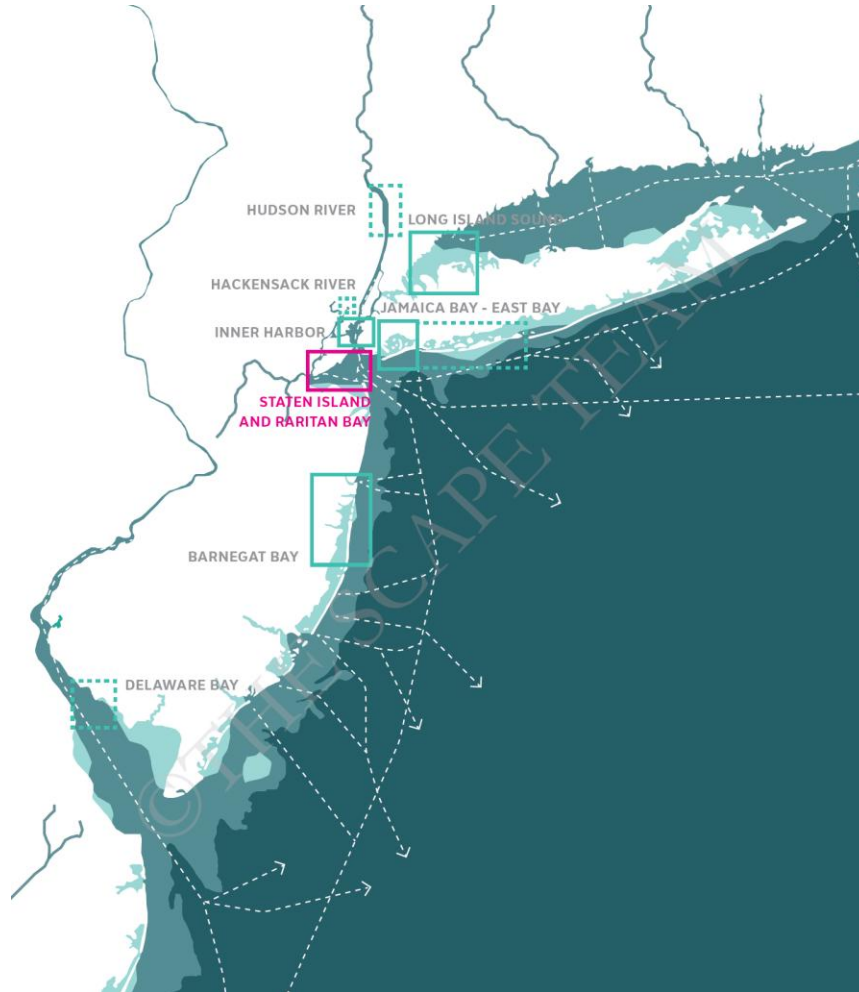
SCAPE Landscape Architecture PLLC. 2014. Living Breakwaters: IP Edition, Staten Island and Raritan Bay. Rebuild By Design. <http://www.rebuildbydesign.org/our-work/all-proposals/winning-projects/ny-living-breakwaters>. Accessed January 4, 2021.

Covered in *Good Morning, America*

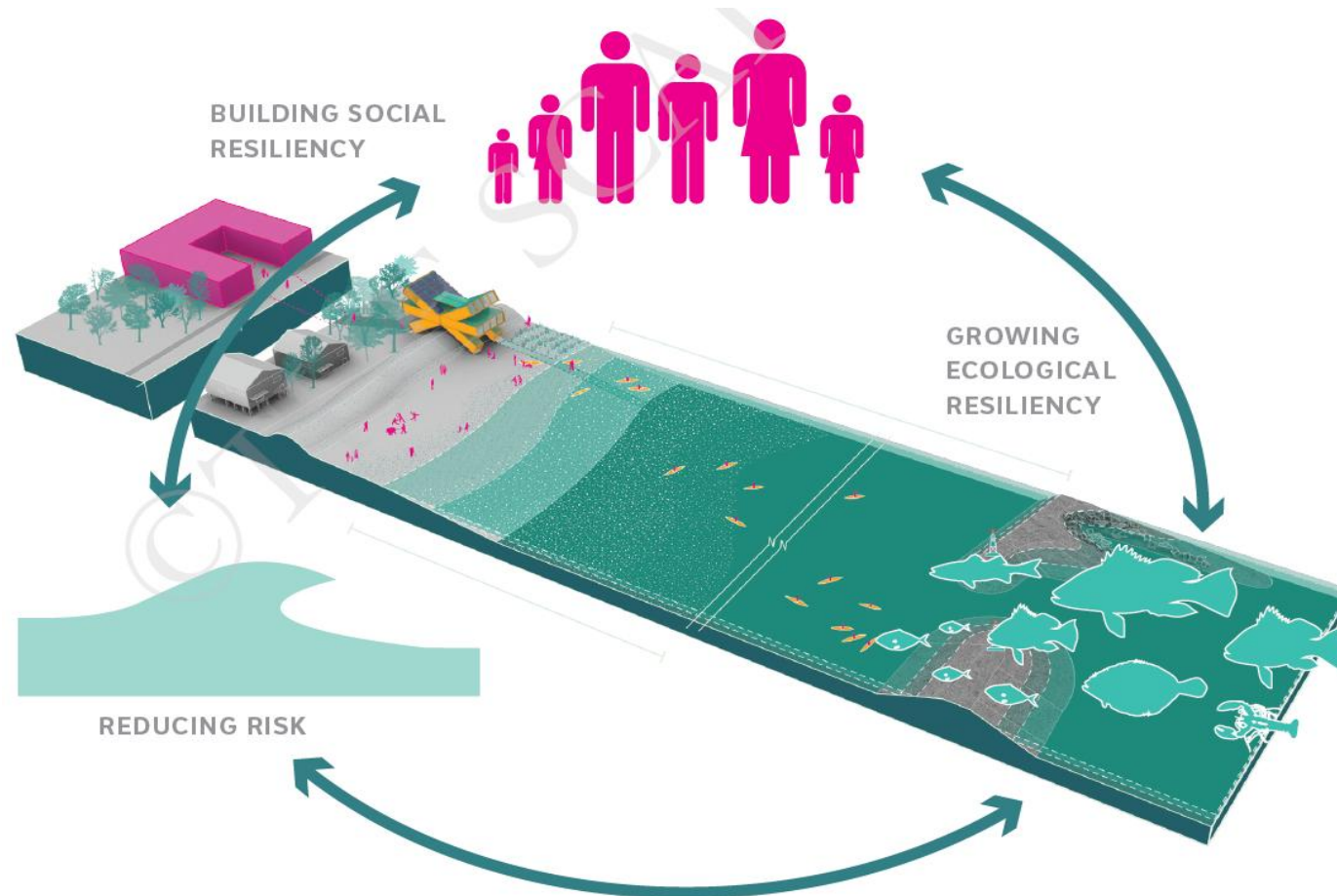
https://www.youtube.com/watch?v=XwH_YCsdLeM



Tottenville, Staten Island: Nested Flood Risk Management



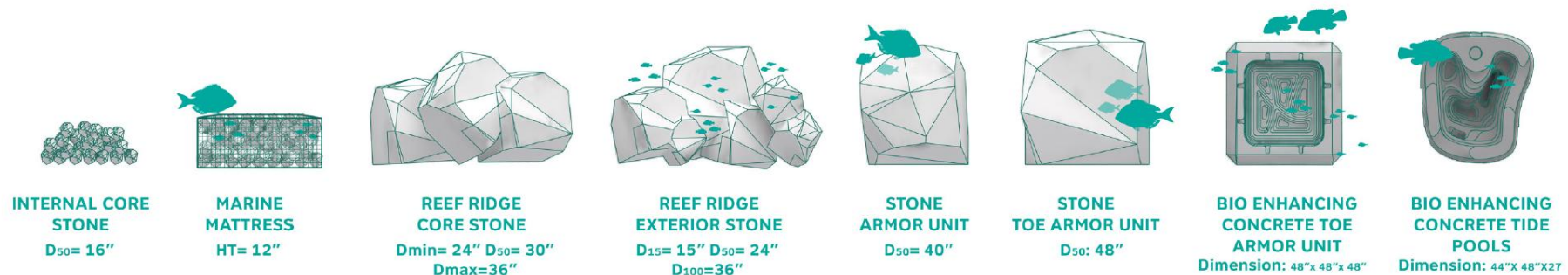
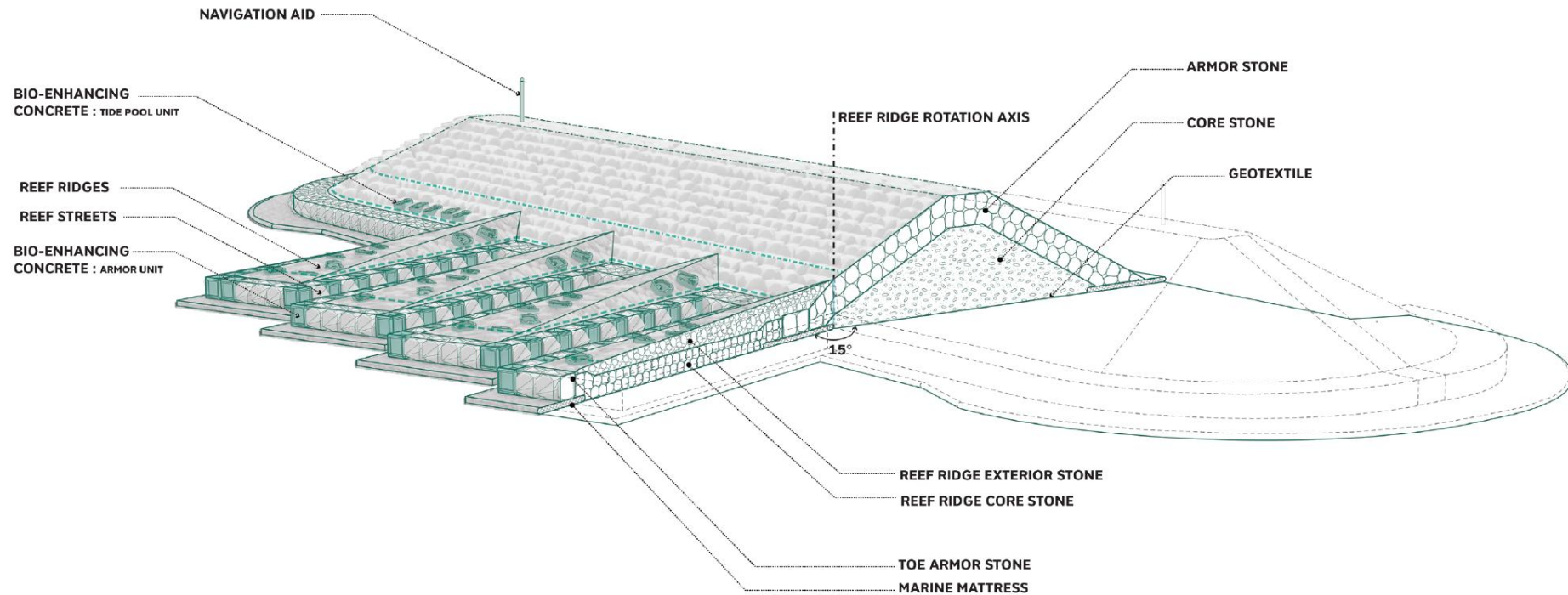
Balancing outcomes in flood risk management



Hybrid shoreline and breakwater systems



Nuanced approach to design



Paper Discussion

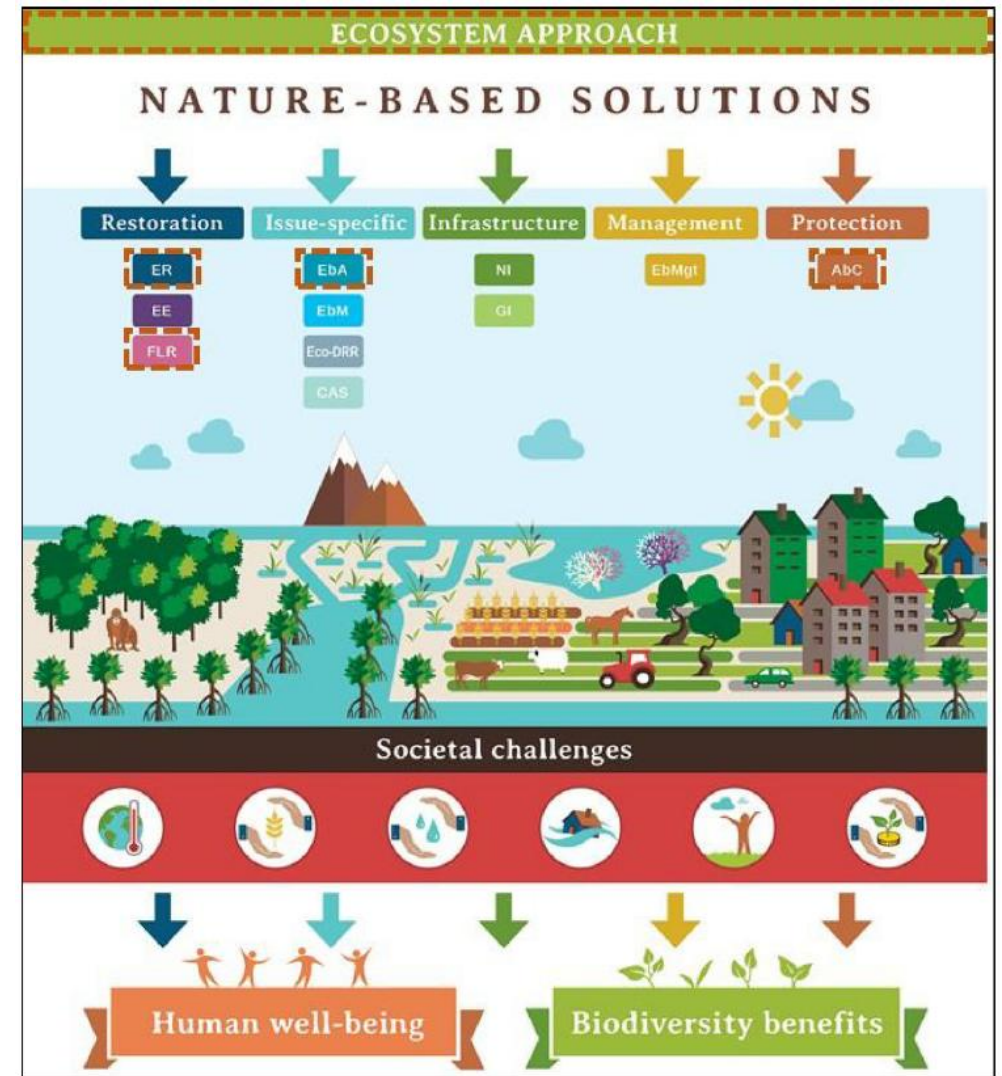
Cohen-Shacham, E., Andrade, A., Dalton, J., Dudley, N., Jones, M., Kumar, C., Maginnis, S., Maynard, S., Nelson, C.R., Renaud, F.G. and Welling, R., 2019. Core principles for successfully implementing and upscaling Nature-based Solutions. *Environmental Science & Policy*, 98, pp.20-29.

Singhvi, A., Luijendijk, A.P. and van Oudenhoven, A.P., 2022. The grey–green spectrum: A review of coastal protection interventions. *Journal of Environmental Management*, 311, p.114824.

Silliman, B.R., Hensel, M.J., Temmink, R.J., Paxton, A.B., Wieczynski, D.J., Gibert, J.P., Johnston, D.W., Noonan, D., Ferris, W.N., Hay, M.E. and Paalman, T., 2026. Ecosystem technology (ecotech): Harnessing natural processes to address global challenges. *Science advances*, 12(19), p.eaec5411.

Cohen-Shacham et al. (2019)

- Should NBS principles overlap with ecosystem restoration? Aren't they different?
- Defining NBS based on what it is not
- Do NBS and natural infrastructure imply different scales (i.e., global and local, space and place)?



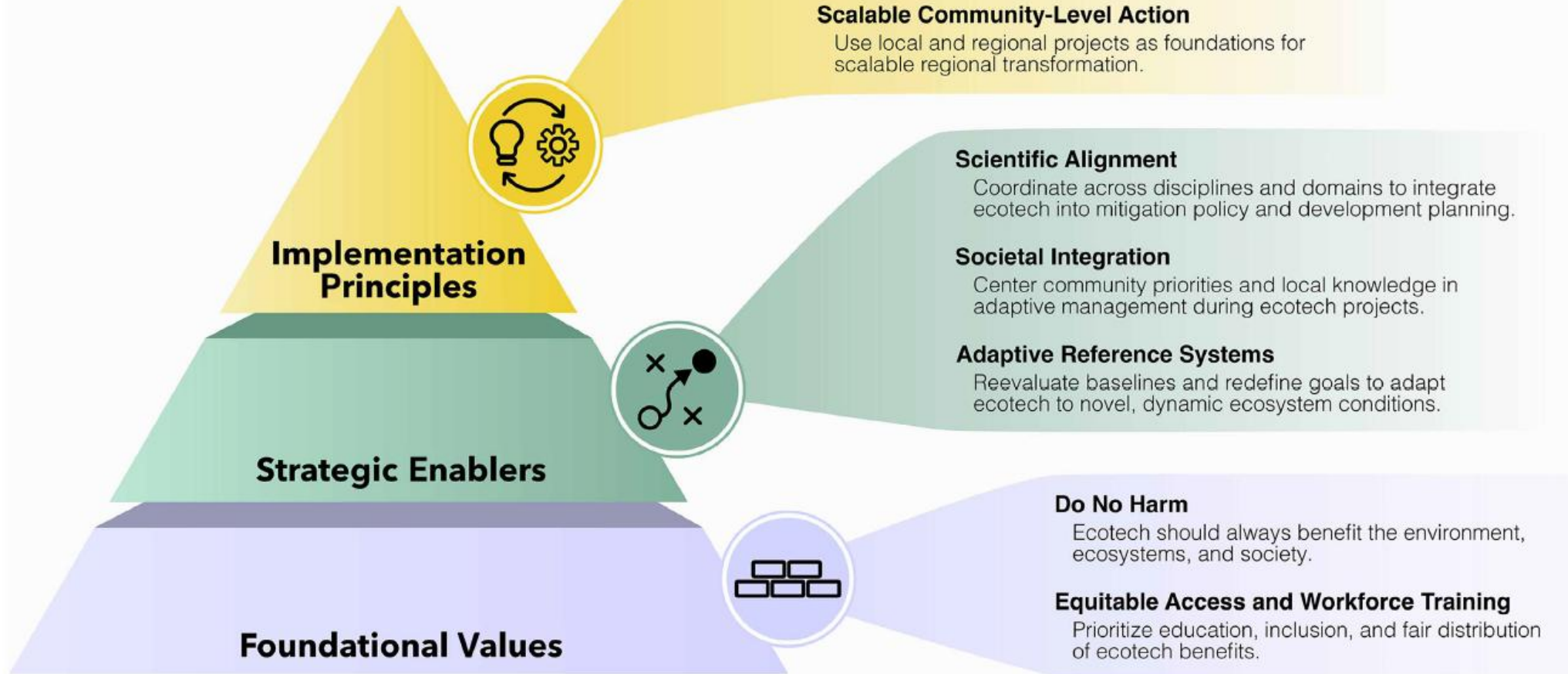
Singhvi et al. (2022)

- Emergent properties (i.e., “internal factors”) are often what makes hybrid and nature-based solutions more appealing.
- “No intervention can eliminate all risk, and taking a holistic perspective showcases the meaningful contributions green and hybrid interventions can make to coastal resilience.”
- “Characteristics that are generally attributed to green interventions were not found to be unique to the green side of the spectrum”
- “Our review suggests that resilient solutions for adaptation are unlikely to be exclusively engineered or natural, but rather a diverse mix of options.”

Silliman et al. (2026)

- The principles of ecotech have strong overlap with NBS. What are the similarities and differences?
- The authors identify three pathways to the market (IP protections, regulatory streamlining, offtakes, and market conditions). Are these unique to ecotech?
- The authors mention off-site and indirect benefits as a major hinderance for ecotech. How can these be assessed?
- If you were a venture capitalist, would you be ready to invest in ecotech? Is biotech an inspiring field to model after?

Guiding Principles of Ecosystem Technology



General Discussion

- Is NBS being held to higher standards?
 - Cook et al. (2025) -> “An identified need was for more inclusive engagement strategies...”
 - Hall and Van Rees (2026) -> “the ideal advanced in the literature is for NBS to...”
- How do we differentiate terminology?
 - Ecosystem restoration, biomimicry, hybrid infrastructure, ecotechnology

Next Week

Ibrahim, A., Marshall, K., Carmen, E., Blackstock, K.L. and Waylen, K.A., 2025. Raising standards for stakeholder engagement in Nature-based Solutions: Navigating the why, when, who and how. *Environmental Science & Policy*, 163, p.103971.

Crawford, B.A., Maerz, J.C. and Moore, C.T., 2026. Navigating multiple objectives, partners, and jurisdictions: Decision analysis for range-wide conservation planning for an at-risk species. *Conservation Science and Practice*, p.e70291.

Pathak A., Del Angel D.C., and Hale C. 2026. Co-developing nature-based solutions for coastal resilience: A case study from Refugio County, Texas. *Estuaries and Coasts*, 49, 31.